Marked Simplicial Sets

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The content of all propositions, definitions and proofs in this essay come from Lurie's Higher Topos Theory[2], however what I have written is my own work.

1 Introduction

In class, we developed the Joyal model structure on simplicial sets. In this model structure, the weak equivalences are the categorical equivalences, the cofibrations are the monomorphisms ($\overline{\text{Cell}}$) and the fibrations are the categorical fibrations. Proving that this defines a model structure was notable work, some coming from $\text{TFib} = \overline{\text{Cell}}^{\boxslash}$ (Proposition 40.1 in Rezk[1]) and applications of the small object argument. Further, we proved that the Joyal model structure is Cartesian closed (Proposition 40.2 in Rezk[1]).

What we want to do is develop a similar model structure, but in the relative case. That is, given a simplicial set S, we would like to define a Cartesian closed model structure on $\text{sSet}_{/S}$. This turns out to be very difficult, and the natural way to extend the Joyal model structure to $\text{sSet}_{/S}$ runs into problems[2]. To overcome these issues, we need a way to distinguish some morphisms in our simplicial sets, hence we introduce marked simplicial sets. These are just simplicial sets with the extra data of some of the morphisms being marked. Eventually, we endow a slice of the category of marked simplicial sets with the Cartesian model structure, a model structure which earns its name from the fibrant objects being Cartesian fibrations[2]. Actually getting to the level of machinery where we can define this model structure is well beyond our current scope, so our goal is to develop the foundation of this model.

2 Marked Simplicial Sets

Definition 2.1: A *marked simplicial set* is a pair $(X, \mathcal{E})$ with X a simplicial set and $\mathcal{E} \subset X_1$ a set of edges in X which contains every degenerate edge.

We say that an edge of X is marked if it belongs to $\mathcal{E}$. Effectively, all we are doing is flagging the degenerate edges of X, and potentially others.

Definition 2.2: A morphism $f : (X, \mathcal{E}) \to (X', \mathcal{E}')$ of marked simplicial sets is a morphism $f : X \to X'$ of the underlying simplicial sets such that $f(\mathcal{E}) \subset \mathcal{E}'$.

That is, a morphism of marked simplicial sets is just a map between simplicial sets that sends marked edges to marked edges.

Example 2.3: There are two obvious extreme cases. Let $S^\sharp = (S, S_1)$ denote the marked simplicial set where every edge of S is marked and let $S^\flat = (S, s_0(S_0))$ denote the marked simplicial set where only degenerate edges of S are marked. There is a natural inclusion $S^\flat \to S^\sharp$ given by the identity morphism $S \to S$ that simply marks all the edges in $S^\flat$. Further, maps $S^\flat \to (M, \mathcal{E}_M)$ are in exact correspondence with maps $S \to M$ since any map of simplicial sets takes degenerate edges of S to degenerate edges of M. In the same way, maps $(S, \mathcal{E}_S) \to M^\sharp$ are in exact correspondence with maps $S \to M$ of simplicial sets.

Example 2.4: For a non trivial example of a marked simplicial set, let $X^\natural = (X, \mathcal{E})$ where $\mathcal{E}$ is the set of p-Cartesian edges of X. The definition of a p-Cartesian edge is given in 3.1 and will be fully explained in that section

We will denote the category of marked simplicial sets as sSet^+. Further, given a simplicial set S, we will write $\mathrm{sSet}^+_{/S}$ to denote the slice category over $S^\sharp$. That is, objects of the form $p : (X, \mathcal{E}_X) \to S^\sharp$ in $\mathrm{sSet}^+_{/S}$ exactly correspond to maps $p : X \to S$ because by marking all edges in X, the condition $f(\mathcal{E}_X) \subset \mathcal{E}_{S^\sharp}$ is trivially satisfied. This means that $\mathrm{sSet}^+_{/S}$ is an enlarged version of $\mathrm{sSet}_{/S}$, and we have freedom to choose our domain $(X, \mathcal{E}_X)$ that has sufficiently nice edges marked, without changing the underlying map $p : X \to S$. This greatly helps us define a model structure on sSet^+, and the edges that we end up choosing to mark are the p-Cartesian edges of X.

Next, we wish to define the class of marked anodyne morphisms in sSet^+, which will be analogous to the class of inner anodyne maps in sSet. Following this, we want to study the class of maps that lift against marked anodyne maps, because this class will be analogous to inner fibrations in sSet and we will call them marked inner fibrations. In order to do this, we first need to define Cartesian fibrations and p-Cartesian edges of a simplicial set.

3 Cartesian Fibrations

Let $p : X \to S$ be an inner fibration of simplicial sets and $j : \Delta^{\{1\}} \to \Delta^1$ be the inclusion of the last vertex of Δ^1 into Δ^1.

Definition 3.1: Let $f : \Delta^1 \to X$ be an edge $x \to y$ in X. Then f is *p-Cartesian* if the pullback slice map

$$p^{\boxtimes_f j} : X_{/f} \to X_{/y} \times_{S_{/p(y)}} S_{/p(f)}$$

is a trivial fibration.

To unpack this definition, first we realise that we are taking the pullback slice of the below diagram.

$$\Delta^{\{1\}} \xrightarrow{j} \Delta^1 \xrightarrow{f} X \xrightarrow{p} S$$

This corresponds to the following pullback diagram.

$$\begin{tikzcd} X_{/f} \arrow[dr, dashed, "p^{\boxtimes_f j}"] \arrow[drr, bend left, "p"] \arrow[ddr, bend right] & & \\ & X_{/y} \times_{S_{/p(y)}} S_{/p(f)} \arrow[r] \arrow[d] & S_{/p(f)} \arrow[d] \\ & X_{/y} \arrow[r, "p"'] & S_{/p(y)} \end{tikzcd}$$

The map $p^{\boxtimes_f j}$ is a trivial fibration if and only if it lifts against all boundary inclusions. Further, for any n, we have an adjoint lifting problem (see Rezk[1], Proposition 26.9).

$$\begin{tikzcd} \partial\Delta^n \arrow[r] \arrow[d, tail, "i"'] & X_{/f} \arrow[d, "p^{\boxtimes_f j}"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & X_{/y} \times_{S_{/p(y)}} S_{/p(f)} \end{tikzcd} \quad \Leftrightarrow \quad \begin{tikzcd} \emptyset \star \Delta^1 \arrow[d, tail] \arrow[dr, "f"] & \\ (\partial\Delta^n \star \Delta^1) \cup_{\partial\Delta^n \star \Delta^{\{1\}}} (\partial\Delta^n \star \Delta^{\{1\}}) \arrow[r] \arrow[d, tail, "i \boxtimes j"'] & X \arrow[d, "p"] \\ \partial\Delta^n \star \Delta^1 \arrow[r] \arrow[ur, dashed] & S \end{tikzcd}$$

The map $i \boxtimes j$ is the pushout product $(\partial\Delta^n \subset \Delta^n) \boxtimes (\Lambda^1_1 \subset \Delta^1)$ which is isomorphic to the inclusion $\Lambda^{n+2}_{n+2} \subset \Delta^{n+2}$ (see Rezk[1], Example 26.3). Thus, we make the following conclusion.

Lemma 3.2: The map $p^{\boxtimes_f j}$ is a trivial fibration if and only if for all $n \geq 2$, the below diagram admits a lift.

$$\begin{tikzcd} \Delta^{\{n-1,n\}} \arrow[d, tail] \arrow[dr, "f"] & \\ \Lambda^n_n \arrow[r] \arrow[d, tail] & X \arrow[d, "p"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & S \end{tikzcd}$$

This gives us a more useful definition of p-Cartesian edges and it is the definition that we will be using most often.

Next, we have two propositions from Higher Topos Theory[2] section 2.4.1 that we will need later. The first is a proposition also covered in Rezk.

Proposition 3.3 (Lurie, HTT 2.4.1.5)**:** Let $p : C \to D$ be an inner fibration of between quasicategories and let $f : c \to c'$ be a morphism in C. Then the following are equivalent:

1. The edge f is an equivalence[1] in C.
2. The edge f is p-Cartesian, and $p(f)$ is an equivalence in D.

Proof. This is exactly the statement of the relative case of Joyal lifting (see Rezk[1], Theorem 28.13). We have essentially shown that if $p(f)$ is an equivalence in D, then the p-Cartesian edges of C are exactly the equivalences in C. □

Proposition 3.4 (Lurie, HTT 2.4.1.7)**:** Let $p : C \to D$ be an inner fibration between quasicategories and let $\sigma : \Delta^2 \to C$ be a 2-simplex of C described by the following diagram.

$$\begin{tikzcd} & c_1 \arrow[dr, "g"] & \\ c_0 \arrow[ur, "f"] \arrow[rr, "h"'] \arrow[rr, phantom, bend left=40, "\sigma"] & & c_2 \end{tikzcd}$$

Suppose that g is p-Cartesian. Then f is p-Cartesian if and only if h is p-Cartesian.

We will leave this proposition without proof in the interest of time and staying on topic. The proof is quite involved and would require unravelling the proofs of several more propositions in Higher Topos Theory.

Finally, we define Cartesian fibrations between simplicial sets.

Definition 3.5: A map $p : X \to S$ of simplicial sets is a *Cartesian fibration* if the following hold:

1. The map p is an inner fibration
2. For every edge $f : x \to y$ of S and every vertex $\tilde{y}$ of X with $p(\tilde{y}) = y$, there exists a p-Cartesian edge $\tilde{f} : \tilde{x} \to \tilde{y}$ with $p(\tilde{f}) = f$.

Similarly, p is a *coCartesian fibration* if $p^{\mathrm{op}} : X^{\mathrm{op}} \to S^{\mathrm{op}}$ is a Cartesian fibration.

4 Marked Anodyne Morphisms

Now we define the saturated class of marked anodyne morphisms in such a way that the class of maps that have the right lifting property against them are 'nice' Cartesian fibrations.

[1] What Lurie calls equivalences, Rezk refers to as isomorphisms and pre-isomorphisms when not necessarily in a quasicategory.

Definition 4.1: The class of *marked anodyne morphisms* in sSet is the saturated class generated by:

(1) Inclusions $(\Lambda^n_i)^\flat \to (\Delta^n)^\flat$ for all $0 < i < n$.

(2) For all $n > 0$, the inclusion $(\Lambda^n_n, \mathcal{E} \cap (\Lambda^n_n)_1) \to (\Delta^n, \mathcal{E})$ where $\mathcal{E}$ is the set of degenerate edges of Δ^n and the final edge $\Delta^{\{n-1,n\}}$.

(3) The inclusion $(\Lambda^2_1)^\sharp \coprod_{(\Lambda^2_1)^\flat} (\Delta^2)^\flat \to (\Delta^2)^\sharp$.

(4) For any Kan complex K, the map $K^\flat \to K^\sharp$.

We denoted the class of marked anodyne morphisms as $\mathrm{InnAnodyne}^+$ and the right complement of these maps as $\mathrm{InnFib}^+ = (\mathrm{InnAnodyne}^+)^\boxslash$, which we call the marked inner fibrations.

At first sight, these definitions look really strange and unorthodox, but we will describe the intuition behind them now and later it will become more clear why these are the maps we want. To understand this class of maps, suppose $p : (X, \mathcal{E}_X) \to (S, \mathcal{E}_S)$ is an element of InnFib^+. Then condition (1) above enforces that p is a trivial fibration.

$$\begin{tikzcd} \Lambda^n_i \arrow[r] \arrow[d, hook] & X \arrow[d, "p"] \\ \Delta^n \arrow[r] \arrow[ur, dashed] & S \end{tikzcd} \quad \Leftrightarrow \quad \begin{tikzcd} (\Lambda^n_i)^\flat \arrow[r] \arrow[d, hook] & X \arrow[d, "p"] \\ (\Delta^n)^\flat \arrow[r] \arrow[ur, dashed] & S \end{tikzcd}$$

As we have mentioned before, maps out of a flat marked simplicial set just correspond to maps between the underlying simplicial sets. So p lifts against inner horn inclusions and therefore p is an inner fibration. Condition (2) is a statement about certain edges being p-Cartesian, which will be made precise in Proposition 4.3.

For condition (3), a map from $(\Lambda^2_1)^\sharp \coprod_{(\Lambda^2_1)^\flat} (\Delta^2)^\flat$ to X is a 2-simplex in X with two edges marked (specifically, the edges of Λ^2_1). A map from $(\Delta^2)^\sharp$ to S is a 2-simplex in S with all edges marked. So the existence of a lift is the statement that the third edge of the 2-simplex in X is marked. This condition is depicted below where a 2-simplex in X is being mapped by p to a 2-simplex in S and the red edges indicate marked edges.

$$\left(\begin{tikzcd} & y \arrow[dr, red] & \\ x \arrow[ur, red] \arrow[rr, "f"] & \sigma & z \end{tikzcd} \quad \overset{p}{\longmapsto} \quad \begin{tikzcd} & p(y) \arrow[dr, red] & \\ p(x) \arrow[ur, red] \arrow[rr, red, "p(f)"] & p(\sigma) & p(z) \end{tikzcd} \right) \quad \Rightarrow \quad f \text{ is marked}$$

Condition (4) is similar to condition (3), but applies to images of Kan complexes in X and S.

Remark 4.2: Generating InnAnodyne$^+$ over every Kan complex could potentially lead to InnAnodyne$^+$ being very big, maybe too large to be a set. However, it is sufficient to allow K to only run over isomorphism classes of Kan complexes with countably many simplices[2]. Therefore, InnAnodyne$^+$ is a set and also is small enough to use the small object argument on.

Now that we have defined InnAnodyne$^+$, we want to understand InnFib$^+$. So we give a necessary and sufficient condition to have the right lifting property against InnAnodyne$^+$.

Proposition 4.3 (Lurie, HTT 3.1.1.6)**:** A map $p : (X, \mathcal{E}_X) \to (S, \mathcal{E}_S)$ in sSet$^+$ is a marked trivial fibration if and only if the following are satisfied:

(A): The map p is an inner fibration (of simplicial sets).

(B): An edge e of X is marked if and only if $p(e)$ is marked and e is p-Cartesian.

(C): For every object y of X and every marked edge $\overline{e} : \overline{x} \to p(y)$ in S, there exists a marked edge $e : x \to y$ of X with $p(e) = \overline{e}$.

Proof. $(\Rightarrow)$: Suppose $p \in$ InnFib$^+$. We need to prove that conditions $(A), (B)$ and (C) hold. We have already shown that p is an inner fibration which proves (A). For (C), let $y \in X_0$ and let $\overline{e} : \overline{x} \to p(y)$ be a marked edge in S. Since $\overline{e}$ is marked, we have a map $(\Delta^1)^\sharp \to (S, \mathcal{E}_S)$ that represents $\overline{e}$.

$$\begin{tikzcd}
(\Lambda^1_1)^\flat \arrow[r, "y"] \arrow[d, hook] & (X, \mathcal{E}_X) \arrow[d, "p"] \\
(\Delta^1)^\sharp \arrow[r, "\overline{e}"'] \arrow[ur, dashed, "e"] & (S, \mathcal{E}_S)
\end{tikzcd}$$

Applying (2) in Definition 4.1, we obtain a lift e, which represents a marked edge in X. This satisfies $p(e) = \overline{e}$ by the commutativity of the diagram, hence this is the required edge and we have proved (C).

To prove the forwards direction of (B), suppose e is a marked edge in X. Then $p(e)$ is marked automatically because p is a map of marked simplicial sets. To show e is p-Cartesian, we are going to use our alternate definition of p-Cartesian edges (Lemma 3.2) and find a lift for the below diagram (left) for any $n \geq 2$.

$$\begin{tikzcd}
\Delta^{\{n-1,n\}} \arrow[d, hook] \arrow[dr, "e"] & \\
\Lambda^n_n \arrow[r] \arrow[d, hook] & X \arrow[d, "p"] \\
\Delta^n \arrow[r] & S
\end{tikzcd}
\quad \Rightarrow \quad
\begin{tikzcd}
(\Delta^{\{n-1,n\}})^\sharp \arrow[d, hook] \arrow[dr, "e"] & \\
(\Lambda^n_n, \mathcal{E} \cap (\Lambda^n_n)_1) \arrow[r] \arrow[d, hook] & (X, \mathcal{E}_X) \arrow[d, "p"] \\
(\Delta^n, \mathcal{E}) \arrow[r] \arrow[ur, dashed] & (S, \mathcal{E}_S)
\end{tikzcd}$$

Since e is a marked edge of X, we have a map $(\Delta^{\{n-1,n\}})^\sharp \xrightarrow{e} (X, \mathcal{E}_X)$. Therefore, we can use the maps from the left to form the diagram on the right and apply (2) to obtain a lift. Taking the underlying map of simplicial sets in this lift, we have a lift for the diagram on the left. That is, we have that e is p-Cartesian and this proves the forwards direction of (B).

Now suppose that $e : x \to y$ is p-Cartesian and $p(e)$ is marked in S. We need to show that e is marked. Applying (C) (which we have already proved), there exists a marked edge $e' : x' \to y'$ with $p(e) = p(e')$. Applying the forwards direction of (B) (which we just proved), we have that e' is p-Cartesian because e' is marked. Consider the below 2-cells.

$$\tau = \begin{tikzcd} & x' \arrow[dr, red, "e'"] & \\ x \arrow[rr, "e"] & & y' \end{tikzcd} \quad , \qquad \gamma = \begin{tikzcd} & p(x) \arrow[dr, red, "p(e)"] \arrow[d, phantom, "s_0p(e)"] & \\ p(x) \arrow[ur, red, "\mathrm{id}"] \arrow[rr, red, "p(e)"] & & p(z) \end{tikzcd}$$

Using the fact that e' is p-Cartesian, we can find a lift σ in the below diagram that fills in τ.

$$\begin{tikzcd} \Delta^{\{n-1,n\}} \arrow[d, hook] \arrow[dr, "e'"] & \\ \Lambda^n_n \arrow[r, "\tau"] \arrow[d, hook] & X \arrow[d, "p"] \\ \Delta^n \arrow[r, "\gamma"'] \arrow[ur, dashed, "\sigma"] & S \end{tikzcd}$$

Thus we have the below 2-cells.

$$\begin{tikzcd} & x' \arrow[dr, red, "e'"] \arrow[d, phantom, "\sigma"] & \\ x \arrow[ur, "d_2\sigma"] \arrow[rr, "e"] & & y' \end{tikzcd} \overset{p}{\longmapsto} \begin{tikzcd} & p(x) \arrow[dr, red, "p(e)"] \arrow[d, phantom, "p(\sigma)"] & \\ p(x) \arrow[ur, red, "\mathrm{id}"] \arrow[rr, red, "p(e)"] & & p(z) \end{tikzcd}$$

Therefore it is enough to show that $d_2\sigma$ is marked because if $d_2\sigma$ is marked, then by (3) we have that e is marked.

Notice that because $p(x) = p(x')$, we have that x, x' are both in $X_{p(x)}$, the fibre above $p(x)$. Fibres of inner fibrations are quasicategories, so $X_{p(x)}$ is a quasicategory. Let $p' : X_{p(x)} \to \{p(x)\}$ be the pullback of p. Since e' and e are p-Cartesian, we have that $d_2\sigma$ is p'-Cartesian in $X_{p(x)}$ by Proposition 3.4. Further, $p'(d_2\sigma) = \mathrm{id}_{p(x)}$ which is an equivalence. Therefore $d_2\sigma$ is an equivalence in $X_{p(x)}$ by Proposition 3.3. That is, we have shown that $d_2\sigma \in K$ where $K = X^{\mathrm{Core}}_{p(x)}$. Applying

(4), we have a lift in the below diagram.

$$\begin{tikzcd} K^\flat \arrow[r, tail] \arrow[d, tail] & (X,\mathcal{E}_X) \arrow[d, "p"] \\ K^\sharp \arrow[r] \arrow[ur, dashed] & (S,\mathcal{E}_S) \end{tikzcd}$$

The existence of this lift means that $K_1 \subset \mathcal{E}_X$. That is, every edge of K is marked in X. But $d_2\sigma \in K$, therefore $d_2\sigma$ is marked in X. Thus e is marked by (3) and this concludes the proof of (B).

$(\Leftarrow)$: Now suppose that conditions $(A), (B)$ and (C) hold. We need to show that the four types of maps described in Definition 4.1 lift against p. From (A), p is an inner fibration and so (1) is satisfied. For (2), we need a lift in the below diagram.

$$\begin{tikzcd} (\Lambda^n_n, \mathcal{E}\cap(\Lambda^n_n)_1) \arrow[r, "g"] \arrow[d, tail] & (X,\mathcal{E}_X) \arrow[d, "p"] \\ (\Delta^n,\mathcal{E}) \arrow[r] & (S,\mathcal{E}_S) \end{tikzcd} \quad \Rightarrow \quad \begin{tikzcd} (\Delta^{\{n-1,n\}})^\sharp \arrow[d, tail] \arrow[dr, "e"] & \\ (\Lambda^n_n, \mathcal{E}\cap(\Lambda^n_n)_1) \arrow[r, "g"] \arrow[d, tail] & (X,\mathcal{E}_X) \arrow[d, "p"] \\ (\Delta^n,\mathcal{E}) \arrow[r] \arrow[ur, dashed] & (S,\mathcal{E}_S) \end{tikzcd}$$

Let $e = g|\Delta^{\{n-1,n\}}$ be the final edge of the outer n-horn represented by g. Then e is marked which implies e is p-Cartesian and so we have a lift for the diagram on the right. Thus maps of the form (2) lift against p.

For (3), we need to show the following implication.

$$\left(\begin{tikzcd} & y \arrow[dr, "f"] & \\ x \arrow[ur, "e"] \arrow[rr, "g"] & \sigma & z \end{tikzcd} \overset{p}{\longmapsto} \begin{tikzcd} & p(y) \arrow[dr, "p(f)"] & \\ p(x) \arrow[ur, "p(e)"] \arrow[rr, "p(g)"] & p(\sigma) & p(z) \end{tikzcd} \right) \Longrightarrow \quad g \text{ is marked}$$

We want to show that g is p-Cartesian by applying Proposition 3.4. However, the only issue is that X and S are not necessarily quasicategories. We claim that we can assume X and S are quasicategories without any loss of generality. To see this, consider the below lifting problems.

$$\begin{tikzcd} (\Lambda^2_1)^\sharp \coprod_{(\Lambda^2_1)^\flat} (\Delta^2)^\flat \arrow[r, "s"] \arrow[d, two heads, "q"'] & (X,\mathcal{E}_X) \arrow[d, "p"] \\ (\Delta^2)^\sharp \arrow[r, "u"] \arrow[ur, dashed, "t"] & (S,\mathcal{E}_S) \end{tikzcd} \quad \Leftrightarrow \quad \begin{tikzcd} (\Lambda^2_1)^\sharp \coprod_{(\Lambda^2_1)^\flat} (\Delta^2)^\flat \arrow[r, "s"] \arrow[d, two heads, "q"'] & (\mathrm{Im}\ s, \mathrm{Im}\ s \cap \mathcal{E}_X) \arrow[d, "p|_{\mathrm{Im}\ s}"] \\ (\Delta^2)^\sharp \arrow[r, "u"] \arrow[ur, dashed, "t"] & (\mathrm{Im}\ u, \mathrm{Im}\ u \cap \mathcal{E}_S) \end{tikzcd}$$

Here, the map q is surjective in the sense that its underlying map of simplicial sets is surjective (it is the identity). Since q is surjective, the image of t must equal the image of s. So these lifting problems are equivalent and we can assume without loss of generality that $X = \text{Im } s$, $S = \text{Im } u$ and $p = p|_{\text{Im } s}$. That is, we can assume without loss of generality that p is a map between quasicategories.

Since e and f are marked, they are p-Cartesian by (B). Applying Proposition 3.4, we have that g is p-Cartesian. Since $p(g)$ is marked and g is p-Cartesian, we have by (B) that g is marked. Therefore (3) lifts against p.

For (4), we need to find a lift in the below diagram.

$$\begin{tikzcd} K^\flat \arrow[r, "s"] \arrow[d, two heads] & X \arrow[d, "p"] \\ K^\sharp \arrow[r, "u"] \arrow[ur, dashed] & S \end{tikzcd}$$

A lift in this diagram is exactly the statement that all edges in Im s are marked (then the lifting map is s). By the same argument as above, we can assume without loss of generality that $X = \text{Im } s$, $S = \text{Im } u$ and therefore p is a map between quasicategories. Let f be an edge in X. By Proposition 3.3, we have that f is an equivalence in X if and only if f is p-Cartesian and $p(f)$ is an equivalence in S. But X and S are Kan complexes because they are images of Kan complexes. Thus, f and $p(f)$ are always equivalences. So we have shown that all $f \in X$ are p-Cartesian. All edges in S are marked because S is the image of $u : K^\sharp \to S$. Therefore $p(f)$ is marked and applying (B), we have that f is marked. That is, all edges of X are marked and (4) lifts against p. □

Using this description of InnFib$^+$, we see that a natural choice of the marked edges of X are the p-Cartesian edges because this leads to the ideal classification of marked inner fibrations.

Corollary 4.4: A map $p : (Y, \mathcal{E}) \to S^\natural$ is a marked inner fibration if and only if the underlying map $p : Y \to S$ is a Cartesian fibration and $(Y, \mathcal{E}) = Y^\natural$.

Proof. We have $(Y, \mathcal{E}) = Y^\natural$ if and only if the marked edges of Y are exactly the p-Cartesian edges of Y which is equivalent to (B). Applying (B), we have that (C) becomes the second condition for p to be a Cartesian fibration in Definition 3.5. Therefore, (A) and (C) are equivalent to p being a Cartesian fibration. □

We have made substantial progress towards endowing sSet$^+$ with the Cartesian model structure. We have defined marked anodyne morphisms, marked inner fibrations and given a necessary and sufficient condition for morphisms to be marked inner fibrations. Further, we have shown that if our simplicial sets are given the natural marked edges structure, then the marked inner fibrations are exactly the Cartesian fibrations. This is extremely useful, because in the Cartesian model structure, we use this to show that the fibrant objects of this model on sSet$^+_{/S}$ are exactly the

Cartesian fibrations over S. Overall, we have achieved our goal of starting to extend the Joyal model structure on sSet to the relative case $\mathrm{sSet}^+_{/S}$.

References

[1] Charles Rezk (2018). Stuff About Quasicategories.

[2] Jacob Lurie (2012). Higher Topos Theory.